$$= \frac{22}{7} \times (1225 - 441)$$

$$= \frac{22}{7} \times 784$$

$$= 22 \times 112 = 2464 \ m^2$$

Example 2.17 The radius of a circular cricket ground is 76 *m*. A drainage 2 *m* wide has to be constructed around the cricket ground for the purpose of draining the rain water. Find the cost of constructing the drainage at the rate of ₹180/- per *sq.m.*

Solution

The radius of the inner circle (cricket ground), $r = 76 \ m$

A drainage is constructed around the cricket ground.

Therefore, the radius of the outer circle, $R = 76 + 2 = 78 \ m$

We have, area of the circular path $= \pi\left(R^2 - r^2\right) \ sq. \ units$

$$= \frac{22}{7} \times (78^2 - 76^2)$$

$$= \frac{22}{7} \times (6084 - 5776)$$

$$= \frac{22}{7} \times 308$$

$$= 22 \times 44 = 968 \ m^2$$

Given, the cost of constructing the drainage per sq.m is ₹180.

Therefore, the cost of constructing the drainage $= 968 \times 180 =$ ₹1,74,240.

Example 2.18 A floor is 10 *m* long and 8 *m* wide. A carpet of size 7 *m* long and 5 *m* wide is laid on the floor. Find the area of the floor that is not covered by the carpet.

Solution

Here, $L = 10 \ m \ \ B = 8 \ m$

Area of the floor $= L \times B$

$$= 10 \times 8$$

$$= 80 \ m^2$$

Area of the carpet $= l \times b$

$$= 7 \times 5$$

$$= 35 \ m^2$$

Therefore, the total area of the floor not covered by the carpet $= 80 - 35$

$$= 45 \ m^2$$

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